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RESEARCH ARTICLE

g-BKMR: Causal Inference for health effects of time-varying correlated environmental mixtures

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Abstract

Exposure to environmental chemicals has been shown to rewire development affecting later health status. Quantifying the joint effect of environmental mixtures over time is crucial to determine intervention timing. However, causal interpretation of longitudinal environmental mixture studies encounters challenges. There is no statistical approach that allows simultaneously for time-varying confounding, flexible modeling, and variable selection when examining the effect of multiple, correlated, and time-varying exposures. To address these gaps, we develop a causal inference method, g-BKMR, which enables us to estimate nonlinear, non-additive effects of time-varying exposures and time-varying confounders, while also allowing for variable selection. An extensive simulation study shows that g-BKMR outperforms approaches that rely on correct model specification or do not account for time-dependent confounding, especially when correlation across time-varying exposures is high or the exposure-outcome relationship is nonlinear. We apply g-BKMR to quantify the contribution of metal mixtures to cardiovascular disease in the Strong Heart Study, a prospective cohort study of American Indians.

KEYWORDS:
Environmental Mixture; time-varying confounding; time-varying exposure; Bayesian Kernel Machine Regression

1 | INTRODUCTION

Exposure to a complex mixture of chemicals and other substances is a pervasive issue. Many of these substances have been linked to adverse health effects, including toxicants and carcinogens such as inorganic arsenic (iAs), which is common in groundwater and certain foods[1]. Evidence has linked metals and air pollution to cardiovascular disease[2], and exposure to iAs has been associated with cardiometabolic risk. Recent studies have also reported associations between As exposure and incident diabetes[3].

[0]**Abbreviations:** ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor

In addition to arsenic, other heavy metals have been linked to cardiovascular disease. For example, higher levels of urinary cadmium are connected with faster rates of increase in blood pressure[4]; high level of selenium exposure is connected with cardiovascular disease.[5] Quantifying the health effect of environmental metal mixtures, and gaining knowledge of their joint effects may drive the formulation of regulations, hence protecting the public's health.

Approaches for the analysis of multiple exposures are now available. Billionnet et al.[6] provided a review of existing methods to analyze the health effects of exposure to multipollutant mixtures. Depending on the question and the setting Weighed Quantile Sum Regression (WQSR)[7], Bayesian Kernel Machine Regression (BKMR)[8], and the Generalized Additive Model (GAM)[9] have been shown to work well in practice. However, these approaches are limited to analyzing data at a single time point, while longitudinal studies that collect data sequentially at multiple time points are common in fields such as epidemiology, sociology, and econometrics. Alternative models that characterize change for longitudinal data, such as Generalized Estimating Equations (GEE) or Generalized Linear Mixed Models (GLMM), do not afford a causal interpretation.

In environmental health sciences, there is a growing demand for approaches that can handle longitudinal data in the presence of multiple correlated exposures. However, there are methodological challenges that lead to uncertainty regarding how to translate the results of environmental mixture studies into policy recommendations. For example, incorrect adjustment for confounding can lead to bias, and most approaches focus on flexible modeling of environmental mixture effects, leaving confounders adjusted assuming a linear relationship[8,9]. This is problematic because confounders may interact with the environmental mixture, leading to biased results. Additionally, a direct confounding adjustment in the outcome model would not be appropriate if the exposure is time-varying[10].

Causal inference methods center on the premise that data from an observational study can be interpreted as originating from a randomized clinical trial in which the treatment was randomly assigned based on (conditional on) measured covariates[11]. Time-varying confounders are factors that have the potential to confound the causal relationship between a time-varying exposure variable and outcome and that themselves change over time and are evaluated frequently throughout the investigation. When the time-varying confounder is itself impacted by the time-varying explanatory variable of interest, traditional approaches (that is, regression adjustment) for addressing confounding cannot be utilized[12]. Approaches to account for time-varying confounding are required, such as g-methods approaches (e.g., standardization or weighting).

Quantifying the joint effect of the environmental mixture over time is crucial to determine intervention timing. However, causal interpretation of longitudinal environmental mixture studies encounters challenges. Currently, there is a lack of statistical approaches that can simultaneously address time-varying confounding, flexible modeling, and variable selection when assessing the effect of multiple, correlated, and time-varying exposures. To bridge this gap in the statistical literature, we have developed a causal inference method, g-BKMR, which allows for the estimation of nonlinear, non-additive effects of both time-varying exposures and confounders while also enabling variable selection. By incorporating methods from a single time point, BKMR,

and the g-formula, g-BKMR offers a unique approach to examine the non-linear and non-additive effects of exposure mixtures when these effects vary over time. With g-BKMR, we can better understand the complex relationship between environmental mixtures and health outcomes, paving the way for more effective public health policies and interventions.

2 | MATERIALS AND METHODS

2.1 | Causal Estimands

2.1.1 | Static Intervention

We consider the setting in which n subjects, labelled $i = 1, ..., n$, enter a study at baseline (time $t = 0$) and are exposed to $A_{t,i} = A_{0,i}$. In a clinical trial, the value of $A_{0,i}$ is determined by the study protocol, usually at random. In an observational study, $A_{0,i}$ is observed and recorded. A collection of covariates, $C_{0,i}$, are also measured at baseline. At the subsequent follow-up visits, the time-varying confounders, $L_{t,i}$ and treatment $A_{t,i}$ are recorded, $t = 1, ..., T$. An outcome Y_i is observed for each subject, measured at the final visit T, Y_i can be continuous or binary.

Along with the *observed outcome* Y_i for each subject, for each possible realization of the treatment history $\bar{a} \in \bar{\mathcal{A}}$, we also define $Y_i^{\bar{a}}$, the *potential outcome* that would have been observed had the subject, possibly contrary to fact, received treatment history $\bar{a}$.

In a simple setting with 3 visits, assume the DAG in Figure 2. 1. Under a static, fixed treatment regime, the overall causal effect of the exposures changing form the 25th percentile to the 75th percentile can be defined as:

$$ACE = E[Y^{A_0=a_0^*, A_1=a_1^*, A_2=a_2^*}] - E[Y^{A_0=a_0, A_1=a_1, A_2=a_2}]$$

where a_0, a_1, a_2 are the 25th percentile of the exposures, a_0^*, a_1^*, a_2^* are the 75th percentile of the exposures.

2.1.2 | Stochastic Intervention

We are also interested in the change in the mean of the outcome for the change of the distributions of the exposures. The Average Causal Effect under a stochastic intervention is defined as

$$ACE = E[Y^{g^*(a_0,a_1,a_2)}] - E[Y^{g(a_0,a_1,a_2)}]$$

where g and g^* are two distributions of the exposures.

In addition to estimating the effect of the metal mixture changing from the 25th percentile to the 75th percentile, we could also estimate the effect of a single metal of the mixture across time points. In addition, we could also be estimating interaction effects.

2.2 | g-formula

The basic idea of g-formula is standardization, a common approach to estimating the expected value of $Y^{\bar{a}}$ in the population.[13] It can be illustrated in a simple time-fixed confounding example with baseline confounders C_0 where the change in the observed mean of the outcome under two different exposure levels is standardized over the observed values c_0 of C_0.

$$E\left(Y^{\bar{a}}\right) = \sum_{C_0} E\left(Y \mid \bar{A} = \bar{a}, C_0 = c_0\right) P\left(C_0 = c_0\right)$$

where the sum is over all possible values c_0 of C_0. The g-formula is a valid estimator for the average causal effect under several identification assumptions.

Using the DAG in Figure 2.1, we now give three identifying assumptions:

1. Consistency:

$$E[Y|A_0 = a_0, A_1 = a_1, A_2 = a_2] = E[Y^{a_0,a_1,a_2}|A_0 = a_0, A_1 = a_1, A_2 = a_2]$$

2. Exchangeability:

- $Y^{a_0,a_1,a_2} \perp\!\!\!\perp A_0 | C_0$
- $Y^{a_0,a_1,a_2} \perp\!\!\!\perp A_1 | C_0, A_0, L_1$
- $Y^{a_0,a_1,a_2} \perp\!\!\!\perp A_2 | C_0, A_0, L_1, A_1, L_2$

3. Positivity:

$$Pr(a_t|a_{t-1}, l_t, c_0) > 0 \text{ for all } t \text{ and all values of } l, c_0 \text{ with } Pr(L = l, C_0 = c_0) \neq 0$$

The general expression of the g-formula for a static fixed intervention when the identifiability assumptions hold is:

$$E[Y^{\bar{a}}] = \sum_{\bar{l}} \mathrm{E}[Y \mid \bar{A} = \bar{a}, \bar{L} = \bar{l}, C_0 = c_0] \prod_{t=0}^{T} f\left(l_t \mid \bar{a}_{t-1}, \bar{l}_{t-1}\right) f(c_0) \tag{1}$$

The positivity assumption is a crucial assumption in the causal inference that states that every individual in the population has a non-zero probability of receiving any level of exposure. In the case of multiple continuous exposures, the violation of this assumption is a common issue because the probability of receiving each exposure level may become very low for certain individuals. The violation of the positivity assumption could lead to an underestimation of the causal effects of the exposure on the outcome.[14] However, for the stochastic intervention approach, the positivity assumption is more likely to be met because this approach uses a simulated intervention that randomizes the exposure distribution, which can be defined ensuring that each individual has a non-zero probability of receiving each exposure level. Therefore, the stochastic intervention approach is a valuable tool for analyzing the causal effects of multiple continuous exposures.

For a stochastic intervention, let $g_0(a \mid C, L) \equiv P_0(A = a \mid W), a \in \mathcal{A}$ denote the conditional distribution of treatment variable A given covariates under the observed data distribution P_0. The positivity assumption needed for the identification of the average treatment effect is given by:

$$\inf_{a \in \mathcal{A}} g_0(a \mid C, L) > 0, \text{ - a.e.}$$

2.3 | Bayesian Kernel Machine Regression

We first review Kernel Machine Regression (KMR) as a framework for estimating the effect of a complex mixture when only a single time point of exposure is measured. For each subject $i = 1, ..., n$, we assume

$$Y_i = h\left(\mathbf{a}_i\right) + \mathbf{c}_i^{\mathrm{T}} \boldsymbol{\beta} + \epsilon_i \tag{2}$$

where $\mathbf{a}_i = \left(a_{i1}, \ldots, a_{iM}\right)^{\mathrm{T}}$ is a vector of M exposure variables. Liu et al.[15] showed that model (2) can be expressed as the mixed model

$$\begin{aligned} y_i &\sim N\left(h_i + \mathbf{c}_i^T \boldsymbol{\beta}, \sigma^2\right) \\ \mathbf{h} &\equiv \left(h_1, \ldots, h_n\right)^{\mathrm{T}} \sim N(\mathbf{0}, \tau \mathbf{K}) \end{aligned} \tag{3}$$

where $\mathbf{K}$ is the kernel matrix, has $(i, j)-$element $K(\mathbf{a}_i, \mathbf{a}_j)$

There are different choices of the kernel. We focus on the Gaussian kernel, which flexibly captures a wide range of underlying functional forms for $h(\cdot)$, although the methods are applicable to a broad choice of kernels. Under model (2.3), we assume

$$\operatorname{cor}\left(h_i, h_j\right) = \exp\left\{-(1/\rho) \sum_{m=1}^{M} \left(a_{im} - a_{jm}\right)^2\right\}$$

where ρ is a tuning parameter that regulates the smoothness of the dose-response function. This assumption implies that two subjects with similar exposures will have more similar risks (a_i close to a_j, and h_i will be close to h_j).

Bobb et al.[8] presented the Bayesian Kernel Machine Regression (BKMR) as a framework to estimate the effect of a complex mixture on a health outcome.

To fit (2), we assume a prior on the coefficients for the confounding variables, $\beta \sim 1$ (flat prior), and $\sigma^{-2} \sim \text{Gamma}\left(a_\sigma, b_\sigma\right)$, where we set $a_\sigma = b_\sigma = 0.001$. We can parameterize BKMR by $\lambda = \tau\sigma^{-2}$, and we assume a Gamma prior distribution for the variance component of λ. For the smoothness parameter ρ, we assume $\rho \sim \text{Unif}(a, b)$ with $a = 0$ and $b = 100$. For additional details regarding BKMR and prior specifications, see Bobb et al.[8,16]

2.4 | Component-wise variable selection

BKMR allows for variable selection within a Bayesian paradigm. Define the augmented Gaussian kernel function as

$$K\left(\mathbf{a},\mathbf{a}';\mathbf{r}\right)=\exp\left\{-\sum_{m=1}^{M}r_m\left(a_m-a'_m\right)^2\right\}$$

where $\mathbf{r}=(r_1,...,r_M)^T$k and define $\mathbf{K}_{\mathbf{a},\mathbf{r}}$ to be the $n\times n$ matrix with (i,j)-element equal to $K(\mathbf{a}_i,\mathbf{a}_j;\mathbf{r})$.

Assuming a "slab-and-spike" prior on the auxiliary parameters,

$$r_m\mid\delta_m\sim\delta_m f_1\left(r_m\right)+\left(1-\delta_m\right)P_0,\quad m=1,\ldots,M$$

$$\delta_m\sim\text{Bernoulli}(\pi)$$

where $f_1(\cdot)$ is a pdf with support on $\mathbb{R}^+$and P_0 denotes the density with point mass at 0. This approach is analogous to Bayesian variable selection approaches for multiple regression problems[17].The posterior mean of the indicator δ_m has the natural interpretation as the posterior probability that component m is an important component of the mixture, or the posterior "inclusion probability" of component m. Other kernel functions may be augmented in a similar way.

2.5 | Hierarchical variable selection

In situations where mixture components are highly correlated, the above formulation that treats components exchangeably may fail because the data may not be able to distinguish among these correlated components. Bobb et al. (2015) proposed a hierarchical variable selection approach that incorporated information on the structure of the mixture into the model:

Suppose that the mixture components $a_1,\ldots,a_M$ can be partitioned into groups $\mathcal{S}_g(g=1,\ldots,G)$ such that within-group correlation is high while across-group correlation is low.

Then assume that the indicator variables from the slab-and-spike prior

$$\boldsymbol{\delta}_{\mathcal{S}_g}\mid\omega_g\sim\text{Multinomial}\left(\omega_g,\pi_{\mathcal{S}_g}\right),\quad g=1,\ldots,G,$$

$$\omega_g\sim\text{Bernoulli}(\pi)$$

where $\delta_{\mathcal{S}_g}=\left(\delta_m\right)_{z_m\in\mathcal{S}_g}$ is the vector of indicator variables and $\pi_{\mathcal{S}_g}$ is the corresponding vector of prior probabilities for the mixture components z_m in group $\mathcal{S}_g$. This approach allows at most a single component from a group (of highly correlated components) to enter into the model at a time.

2.6 | g-BKMR

Assuming the DAG of Figure 2.1, the g-formula can be calculated by equation (2.1). For the g-formula to be implemented, we must therefore estimate $\mathrm{E}[Y \mid \bar{A}=\bar{a}, \bar{L}=\bar{l}, C_0=c_0]$ and $f\left(l_t \mid \bar{a}_{t-1}, \bar{l}_{t-1}\right) f(c_0)$ from the observed data. This involves postulating a model for Y given $\bar{A}$ and $\bar{L}$ and models for L_t given $\bar{A}_{t-1}$ and $\bar{L}_{t-1}$.

g-BKMR is a Bayesian semi-parametric regression method that will allow for (i) estimation of joint effects and interactions of time-varying environmental exposures, (ii) time-varying confounding. Using g-BKMR, we estimate the causal effect for a change in exposure profile from $\mathbf{a}$ to $\mathbf{a}^*$ via the following algorithm.

1. Fit BKMR time-varying confounding and outcome models:

$$L_i = h_L\left(\mathbf{A}_i\right) + \mathbf{C}_i^T \boldsymbol{\beta} + \epsilon_{Li} \tag{4}$$

$$Y_i = h_Y\left(\mathbf{A}_i, \mathbf{L}_i\right) + \mathbf{C}_i^T \boldsymbol{\theta} + \epsilon_{Y_i} \tag{5}$$

2. For each Markov chain Monte Carlo (MCMC) iteration, $j = 1, ..., J$

(a) Generate $k = 1, ...K$ samples of the confounder for the mean level of covariates under exposure level a^* from confounder model(4).

$$\begin{aligned} L_{\mathbf{a}^\star}^{(jk)}(\bar{\mathbf{c}}) &= \mathrm{E}^{(j)}\left(L \mid \mathbf{A}=\mathbf{a}^*, \mathbf{C}=\bar{\mathbf{c}}\right) + \sigma_L^{(j)} N(0,1) \\ &= h_L^{(j)}\left(\mathbf{A}=\mathbf{a}^*\right) + \bar{\mathbf{c}}^T \boldsymbol{\beta}^{(j)} + \sigma_L^{(j)} N(0,1) \end{aligned}$$

(b) For each of the $j = 1, \ldots, J$ iterations and $k = 1, \ldots, K$ samples of $L_{\mathbf{a}^*}$ estimate the average outcome value for the mean level of covariates for $Y_{\mathrm{a}L_{\mathrm{a}^*}}$ from model (5)

$$\begin{aligned} Y_{\mathrm{a}^* L_{a^*}}^{(jk)}(\bar{\mathbf{c}}) &= \mathrm{E}^{(j)}\left(Y \mid \mathbf{A}=\mathbf{a}^*, L=L_{\mathbf{a}^*}^{(jk)}(\bar{\mathbf{c}}), \mathbf{C}=\bar{\mathbf{c}}\right) \\ &= h_Y^{(j)}\left(\mathbf{A}=\mathbf{a}^*, L=L_{\mathbf{a}^*}^{(jk)}(\bar{\mathbf{c}})\right) + \bar{\mathbf{c}}^T \boldsymbol{\theta}^{(j)} \end{aligned}$$

(c) Let the jth posterior sample of $Y_{\mathrm{a}^* L_{a^*}}$ be the mean over the K samples

$$Y_{\mathrm{a}^* L_{\mathrm{a}}^*}^{(j)}(\bar{\mathbf{c}}) = \frac{1}{K} \sum_{k=1}^{K} Y_{\mathrm{a}^* L_i}^{(jk)}(\bar{\mathbf{c}}) = E\left[Y^{a*(j)} \mid \bar{c}\right]$$

3. The potential outcome Y^{a^*} is then estimated as:

$$\frac{1}{J} \sum_{j=1}^{J} Y_{\mathrm{a}^* L_{\mathrm{a}}^*}^{(j)}(\bar{\mathbf{c}})$$

The jth posterior sample of the ATE is obtained by:

$$AT\,E^j = Y_{\mathrm{a}^* L_{a^*}^{(j)}(\bar{\mathbf{c}})} - Y_{\mathrm{a} L_a^{(j)}(\bar{\mathbf{c}})}$$

4. Estimate the 95 % credible intervals from these posterior samples.

3 | SIMULATION STUDY

We evaluate the ability of g-BKMR to estimate the joint effect of a mixture compared to the traditional regression method with direct confounding adjustment in the outcome model, g-formula, g-formula with correct model specification, and BKMR with hierarchical variable selection (grouped by time point), which ignores time-dependent confounding.

3.1 | Setup

In order to generate the true causal effect, we generated a population dataset with 1,000,000 observations, where $\mathbf{A_t} = (A_{t,1}, ..., A_{t,k})^T$ represents the kth metal exposure (e.g. Arsenic) observed at the tth time point. We generated C_0 as baseline covariates and L_t as time-varying confounders. In each simulation, we randomly sample smaller data sets from this population. We assume normal distributions for the time-varying confounders and assume a multivariate normal distribution for log-transformed metal exposures. Scenarios 1-4 consider scenarios with 2 time points, and scenario 5-6 consider scenarios with 3 time points, assuming a scenario with 3 time points, i.e., $t = (0, 1, 2)$.

$$\log A_0 \sim N(h_{A_0}(C_0), \sigma^2_{A_0})$$

$$L_1 \sim N(h_{L_1}(A_0, C_0), \sigma^2_{L_1})$$

$$\log A_1 \sim N(h_{A_1}(A_0, L_1), \sigma^2_{A_1})$$

$$L_2 \sim N(h_{L_1}(A_1, L_1), \sigma^2_{L_1})$$

$$\log A_2 \sim N(h_{A_2}(A_1, L_2), \sigma^2_{A_2})$$

We considered two types of health outcomes: continuous and binary outcomes. The continuous outcome is generated by

$$y_i \sim N(h_Y(L_1, L_2, A_0, A_1, A_2), \sigma^2_Y)$$

The binary outcome is generated by

$$y_i \sim Bin(n, h_Y(L_1, L_2, A_0, A_1, A_2))$$

We set $\sigma^2_{A_t}$, $\sigma^2_{L_t}$, and σ^2_Y to be realistic noise based on the Strong Heart Study application (Section 2.4).

For the true underlying exposure-confounder functions $h_{L_t}(\cdot)$ and exposure-response functions $h_{Y_t}(\cdot)$, we considered six scenarios for the continuous outcome and four scenarios for the binary outcome, respectively. We considered both linear and non-linear

Scenarios	Time Points	Correlation	Confounding	Underlying Relationship	Outcome
1	2	moderate	low	quadratic + interaction	continuous
2	2	moderate	strong	linear	continuous
3	2	moderate	strong	quadratic	continuous
4	2	moderate	low	quadratic + interaction	binary
5	3	moderate	strong	quadratic	binary
6	3	moderate	strong	linear	binary

Table 1 Number of time points, the strength of the confounding effect, and exposure-confounder-response underlying relationships used for each simulation scenario.

scenarios, weak and strong confounding effects, and with or without interaction effects. The specification of each scenario is summarized in Table 2.1.

For each simulation, we randomly sampled 500 datasets of 500 observations from the population dataset. We first fit BKMR models (6), (7), (8):

$$L_{1i} = h_L\left(\mathbf{A}_{0i}\right) + \mathbf{C}_i^T \boldsymbol{\beta} + \epsilon_{L_{1i}} \tag{6}$$

$$L_{2i} = h_L\left(\mathbf{A}_{0i}, \mathbf{A}_{1i}, \mathbf{L}_{1i}\right) + \mathbf{C}_i^T \boldsymbol{\beta} + \epsilon_{L_{2i}} \tag{7}$$

$$Y_i = h_Y\left(\mathbf{A}_{0i}, \mathbf{A}_{1i}, \mathbf{A}_{2i}, \mathbf{L}_{1i}, \mathbf{L}_{2i}\right) + \mathbf{C}_i^T \boldsymbol{\theta} + \epsilon_{Y_i} \tag{8}$$

Using g-BKMR, we estimated the change of potential outcome Y, for a change of the exposure mixture from $\mathbf{a}$, the exposures set equal to their 25th percentile in the true underlying dataset, to $\mathbf{a}^*$, where the exposures set equal to their 75th percentile in the true underlying dataset.

The performance of the following five methods is compared: (1) g-BKMR with variable selection; (2) g-BKMR with no variable selection; (3) a parametric g-formula with correct model specification; (4) traditional regression – Linear or Logistic regression that adjusts for all exposures and confounders at all 3 time points; (5) a parametric g-formula adjusting for all exposures and covariates; (6)BKMR with hierarchical model selection grouped by time points.

3.2 | Simulation Results

Table 2.2 and Table 2.3 summarize the empirical bias, MSE, and coverage probabilities of the estimates of the average treatment effect across the 500 simulated datasets. Figure 2.2 visualizes the results.

The parametric g-formula with correct model specification yields low levels of bias and high coverage. However, in practical applications, the true functional form of the model is unknown, especially when it involves complex relationships, such as quadratic terms or interactions. Hence, while the g-formula approach with a correctly specified model can serve as a useful benchmark in simulation studies, its practical utility in real-world applications may be limited.

The results also show that traditional linear models, such as Linear Regression or Logistic Regression, exhibit high levels of bias, particularly in scenarios where the underlying relationships between variables are nonlinear. This is because traditional linear models fail to account for the time-varying structure of the data, which further exacerbates the bias.

In cases where the underlying relationship is linear, specifically in scenarios 2 and 6, the coverage probabilities for parametric g-formula are decent. However, when the underlying relationship deviates from linearity, as observed in scenarios 1, 3, 4, and 5, the coverage probabilities of the parametric g-formula approach substantially decrease. These findings highlight the heavy reliance of the parametric g-formula on the assumption of accurate model specification. Given that mixture studies often involve non-linear underlying relationships, relying solely on the parametric g-formula may not yield the most reliable results.

g-BKMR consistently exhibits lower bias across most scenarios. Additionally, we observe that g-BKMR produces estimations with levels of bias that are comparable to those generated by the parametric g-formula approach under the condition of correct model specification. However, it is crucial to recognize that the assumption of a correctly specified model is often impractical in real-world applications. Therefore, the observed similarities between g-BKMR and the g-formula approach are promising in light of this practical limitation. In terms of the comparison between g-BKMR with and without variable selection, our results reveal that both methods yield similar levels of bias and coverage probabilities in most cases, this indicates that g-BKMR does not lose much efficiency by implementing the variable selection. g-BKMR without variable selection slightly outperforms g-BKMR in some scenarios. It is important to note, though, that g-BKMR without variable selection demands considerably higher computational power since it computes point estimates for all predictors instead a subset of it. Therefore, if computational time is a factor to consider, we recommend utilizing g-BKMR with variable selection.

BKMR with grouped variable selection exhibits the highest bias and lowest coverage probabilities among all methods, especially when there are strong confounding effects. This is because BKMR ignores the time-varying confounding.

4 | APPLICATION TO STRONG HEART STUDY

4.1 | Study Population

The Strong Heart Study (SHS)[18] is a prospective cohort study of American Indian men and women. The study commenced from 1989 to 1991, recruiting men and women of age 45-75 from American Indian communities based in Arizona, Oklahoma, and South Dakota. Participants reported age, sex, education, smoking history, alcohol use, and medical history in a baseline questionnaire. Measurement of waist circumference, blood pressure, height, and weight, as well as a collection of urine and fasting blood samples, was performed during physical examinations by centrally trained nurses following a standardized protocol.

The study collected biospecimens to evaluate arsenic (As) exposure as well as measures of antimony (Sb), cadmium (Cd), lead (Pb), molybdenum (Mo), selenium (Se), tungsten (W), uranium (U) and zinc (Zn). The metals are measured at baseline

and 2 follow-ups (between 1993 and 1995, and between 1998 and 1999). Urinary metals were measured in 386 participants at baseline and two follow-up visits, among the above 9 metals, uranium was only measured at the two follow-up visits but not the baseline. Other important measurements include glucose, cholesterol, triglycerides, high-density lipoprotein cholesterol, plasma creatinine, fasting insulin, glycated hemoglobin A1C, urine creatinine, and albuminuria.

The primary outcomes of the SHS study are cardiovascular-related events. Blood glucose is the outcome considered in our g-BKMR analyses. The time-varying confounder is waist circumference, the baseline covariates are age, gender, education, and body mass index (BMI). The main exposures of interest include multiple metals (As, Se, Cd, Pb, Mo, Se, W, and U). Studies suggested that diabetes affects zinc homeostasis in many ways and that hyperglycemia is responsible for increased urinary loss and decreases in total body zinc.[19] Hence, Zinc (Zn) was not included in the analysis as the role of Zn in the clinical management of diabetes is unclear.

Figure 2.3 shows the correlation patterns among metals measured during three separate visits. We can observe a strong positive correlation between the same metal across the three time points, as well as a significant association between different metals. These results suggest that the presence of one metal in a sample may be indicative of the presence of other metals as well, highlighting the importance of considering multiple metals when assessing environmental exposures.

The distribution of participant age by gender in the Strong Heart Study is presented in Figure 2.4. The average age among female participants is 54.78; the average age among male participants is 56.39. The results indicate that female participants are, on average, younger than their male counterparts. Furthermore, a greater number of females (n = 264) were included in the study compared with males (n = 123). Since gender may impact the relationship between the metal mixtures and the health outcome, we examine the relationship within each gender while controlling for potential confounding variables. A stratified analysis is warranted to ensure appropriate comparisons are made.

4.2 | Models

We modeled the joint effect of co-exposure to the metal mixtures (As, Se, Cd, Pb, Mo, Se, W, and U) on the blood glucose level at the last time point, while considering waist circumference as the time-varying confounder. To model the joint effect of the metal mixtures at baseline on the waist circumference at the second visit, we utilize a BKMR model including all metals at baseline along with baseline confounders; To model the joint effect of the metal mixtures at first two visits on the waist circumference at the third visit, we fit a BKMR model including all metals from the first two visits, baseline confounders and waist circumference from the previous visit. All models were adjusted for the baseline age, height, BMI, education, baseline waist circumference, and study center, and were stratified by gender to allow for an examination of potential gender-specific effects. We graphically examine the relationship between the metal mixtures and the blood glucose and the relationship between the metal mixtures and the waist circumference at the second and third visit.

We simulated counterfactuals to estimate the ATE for a change in the raw (untransformed) exposures from all metals set at their corresponding 25th percentile, to all metals set at their 75th percentile. The metals are log-transformed and adjusted by urinary creatinine.

4.3 | Results

We found significant harmful effects of the metal mixtures on blood glucose among female participants. On average, the blood glucose increased by 0.63 (95% CI: (0.082, 1.385)) among females for a change in the mixture from the 25th to the 75th percentile across three time points. The top 3 metals with the highest posterior inclusion probabilities are Sb at time 3, As at time 2, and Mo at time 1. On average, the blood glucose decreased by 0.158 (95% CI: (-0.999, 0.537)) among males. The top 3 metals with the highest posterior inclusion probabilities are Mo at time 3, W at time 2, and Pb at time 1.

Figures 2.5 and 2.6 show the relationship between the metal mixtures and waist circumference at the second time point in males and females, suggesting nonlinear associations of the metals and waist circumference.

Figures 2.7 and 2.8 show the relationship between the metal mixtures, waist circumference and the outcome (blood glucose) in males and females, suggesting nonlinear associations of the metals and blood glucose.

5 | DISCUSSION

Environmental exposures to metal mixtures have been shown to increase the risk of several chronic diseases, including diabetes and cardiovascular disease. In this study, we proposed a novel method, g-Bayesian kernel machine regression (g-BKMR), to estimate the joint effect of multiple time-varying exposures over time, while adjusting for time-varying confounders. Our approach allows for the estimation of joint effects and interactions of time-varying environmental exposures, which is critical for understanding the complex relationships between exposures and health outcomes. Moreover, our method accounts for time-varying confounding, which can also be extended to help mitigate potential biases that may arise from unmeasured confounding factors.

The g-formula used in our method allows for complex relationships between the elements of the mixture and the confounders through the joint kernel specification in the outcome model. Importantly, our proposed method can be applied to a wide range of data with high-dimensional time-varying exposures, making it a valuable tool in environmental health research.

In our simulation study, we compared the performance of g-BKMR to traditional linear models g-formula, g-formula with correct model specification and BKMR with grouped variable selection in estimating the average treatment effect (ATE) in the presence of moderate to strong time-varying confounding and complex exposures. Our results showed that g-BKMR outperformed the other methods, with significantly better coverage probability. Therefore, we recommend using g-BKMR in environmental mixture studies with complex exposures and time-varying confounding.

Applying the g-BKMR method to the Strong Heart Study, we found a harmful effect of co-exposure to Arsenic, Cadmium, Lead, Antimony, Molybdenum, Selenium, Tungsten, and Uranium on blood glucose, among female participants. This finding highlights the importance of addressing exposure to metal mixtures in the prevention of diabetes and cardiovascular disease.

Despite the advantages of our proposed method, there are limitations that need to be addressed. As the number of time points, exposures, and sample size increase, the computational power required to fit the BKMR models increases exponentially. Therefore, in future work, we plan to explore ways to jointly model the BKMR models to reduce the computation time needed.

In conclusion, the g-BKMR method provides a powerful and flexible approach to estimate the joint effects of time-varying exposures, while accounting for time-varying confounders, and can be applied to a wide range of data. The findings from our application of g-BKMR to the Strong Heart Study underscore the importance of addressing exposure to metal mixtures in the prevention of chronic diseases. Further research is needed to improve the computational efficiency of the method to allow for its application to larger datasets.

6 | ACKNOWLEDGEMENTS

7 | REFERENCE

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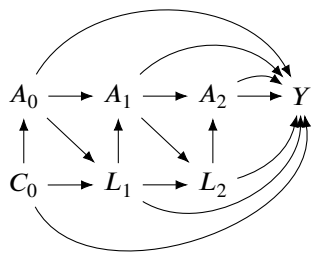

Figure 1 Directed acyclic graphs showing time-varying exposures and time-varying confounders at 3 visits.

Scenario	Methods	Bias	MSE	Coverage
1	gBKMR	0.00	0.01	0.98
1	g-formula	0.26	0.08	0.44
1	Linear Model	0.26	0.08	0.47
1	BKMR_group	-0.02	0.01	0.91
1	gBKMR_noselection	-0.00	0.02	0.99
1	gfromula_correct	0.01	0.01	0.93
2	gBKMR	0.01	0.03	0.96
2	g-formula	-0.14	0.04	0.84
2	Linear Model	-0.14	0.04	0.81
2	BKMR_group	-0.02	0.63	0.01
2	gBKMR_noselection	-0.02	0.04	0.99
2	gfromula_correct	-0.00	0.02	0.95
3	gBKMR	-0.13	0.05	0.91
3	g-formula	0.76	0.64	0.11
3	Linear Model	-0.38	0.25	0.78
3	BKMR_group	1.19	1.47	0.00
3	gBKMR_noselection	-0.11	0.07	0.99
3	gfromula_correct	0.09	0.03	0.95

Table 2 Bias, Mean Square Error (MSE) and coverage probabilities of the ATE estimates from our simulation under varying underlying relationships with a continuous outcome comparing our g-BKMR approach with and without variable selection, the parametric g-formula, traditional linear model, BKMR with grouped variable selection, and g-formula with correct model selection.

Scenario	Methods	Bias	MSE	Coverage
4	gBKMR	0.03	0.00	0.77
4	g-formula	0.03	0.00	0.80
4	Linear Model	-0.25	0.09	0.68
4	BKMR_group	0.03	0.00	0.81
4	gBKMR_noselection	0.01	0.00	0.99
4	gfromula_correct	-0.03	0.00	0.83
5	gBKMR	-0.02	0.00	0.93
5	g-formula	0.08	0.14	0.78
5	Linear Model	0.07	0.01	0.93
5	BKMR_group	0.08	0.01	0.45
5	gBKMR_noselection	0.01	0.00	0.97
5	gfromula_correct	-0.02	0.00	0.92
6	gBKMR	-0.02	0.00	0.94
6	g-formula	-0.10	0.02	0.92
6	Linear Model	0.04	0.01	0.95
6	BKMR_group	0.07	0.01	0.79
6	gBKMR_noselection	0.00	0.00	0.98
6	gfromula_correct	-0.02	0.00	0.93

Table 3 Bias, Mean Square Error (MSE) and coverage probabilities of the ATE estimates from our simulation under varying underlying relationships with a binary outcome comparing our g-BKMR approach with and without variable selection, the parametric g-formula, traditional linear model, BKMR with grouped selection, and g-formula with correct model selection.

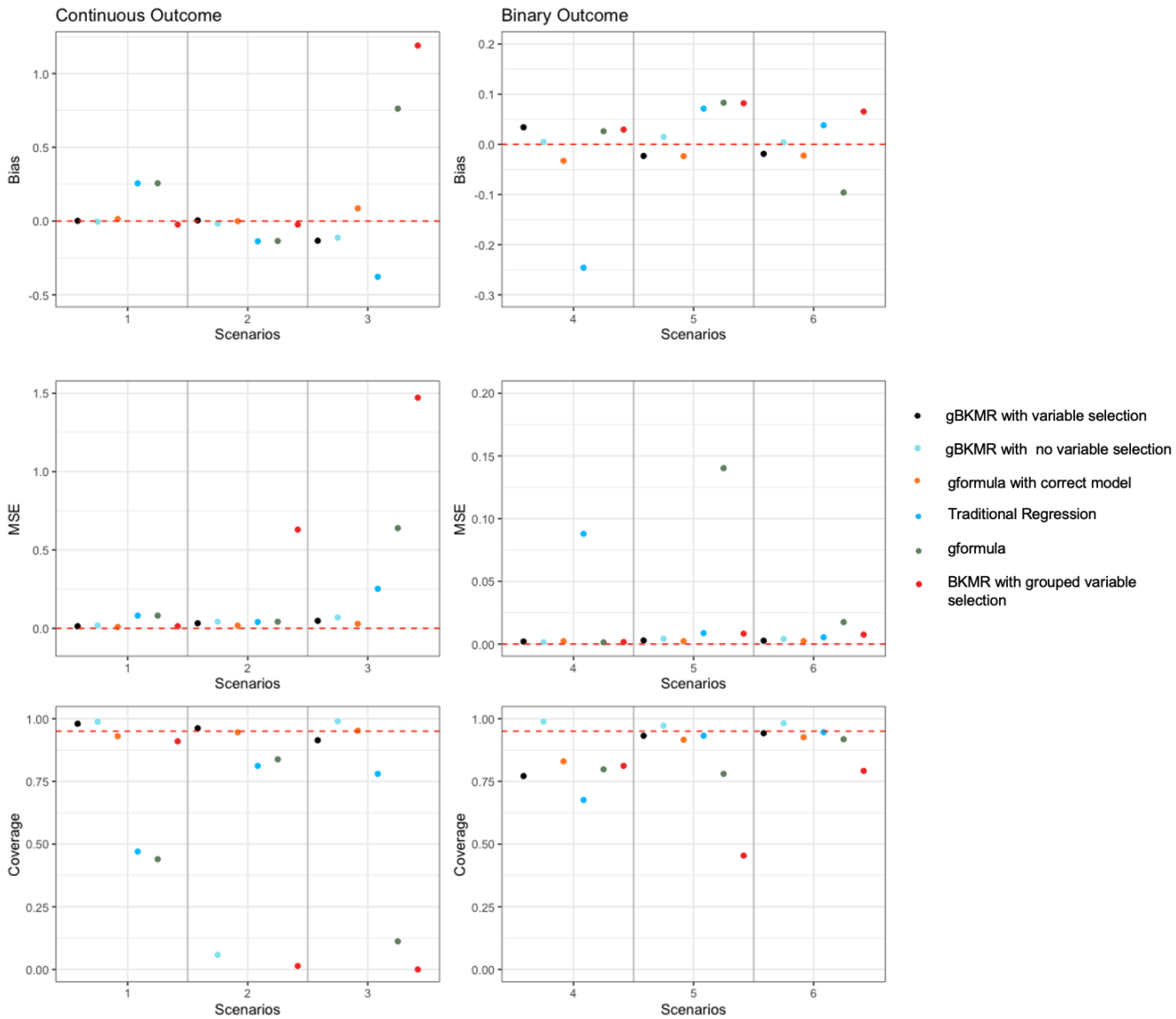

Figure 2 Bias, MSE, and Coverage for simulated scenarios

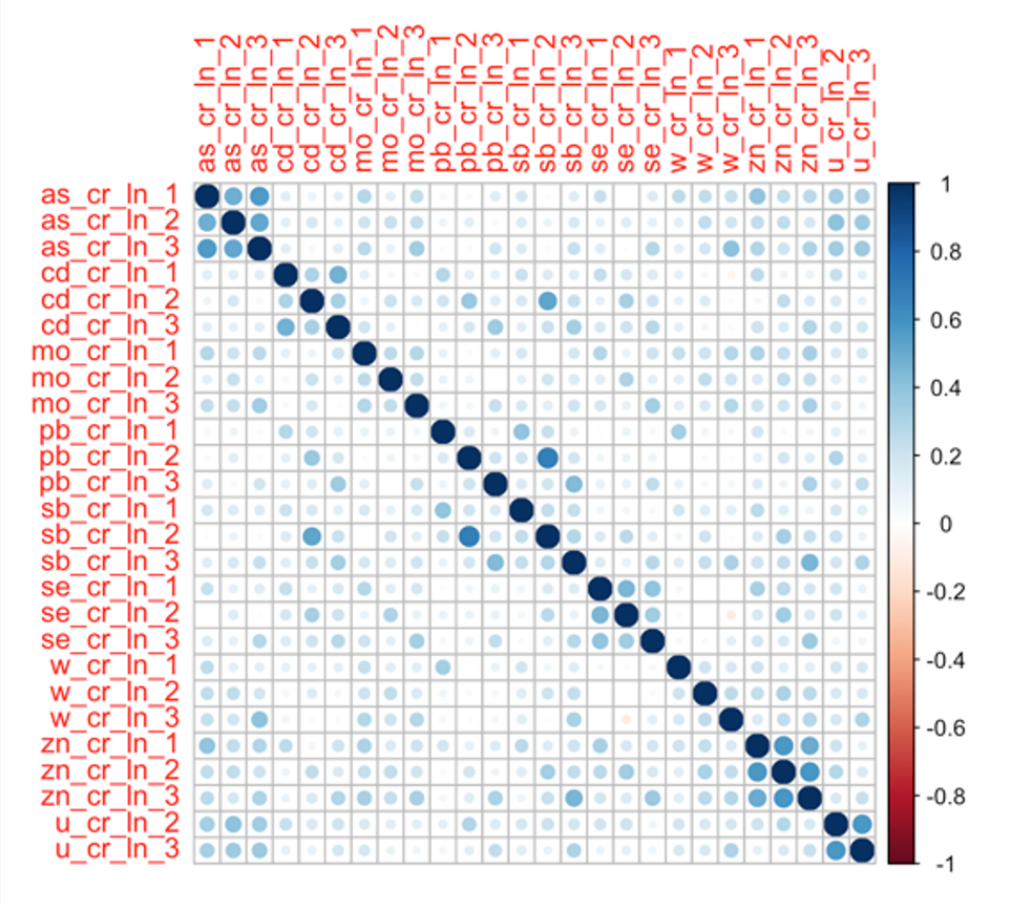

Figure 3 Correlation between metal mixtures at 3 time points in the Strong Heart Study Data

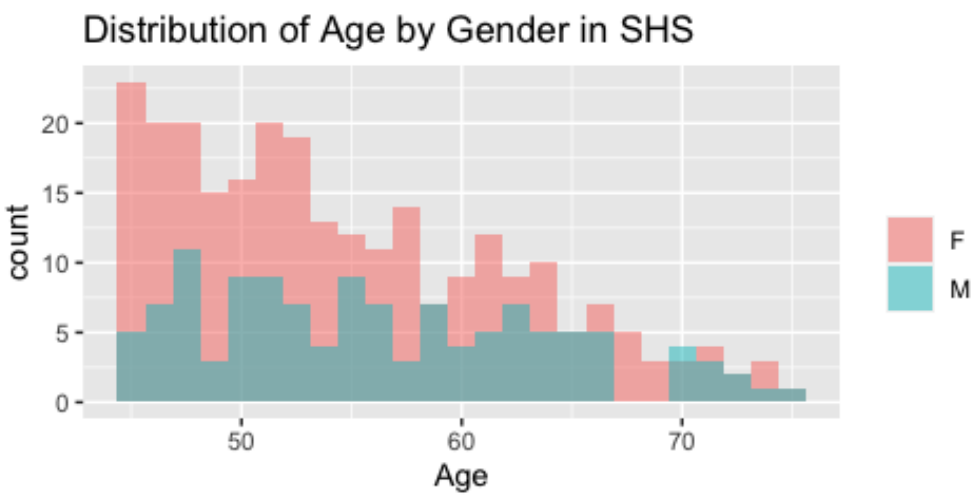

Figure 4 Correlation between metal mixtures at 3 time points in the Strong Heart Study Data

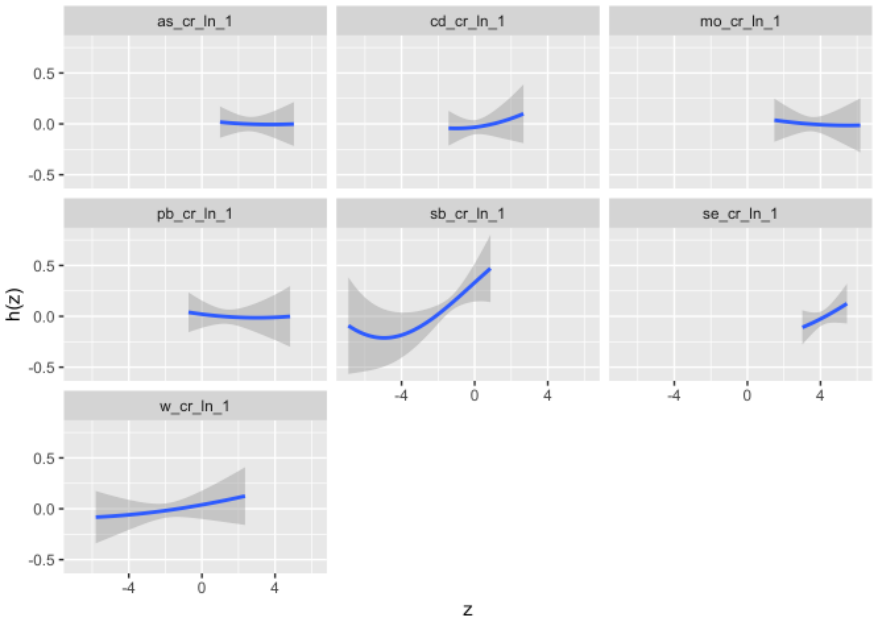

Figure 5 Predictor-response function for the first time-varying confounder, compares the increase in waist circumference at the second visit when a single pollutant is at the 75th versus 25th percentile, for female participants

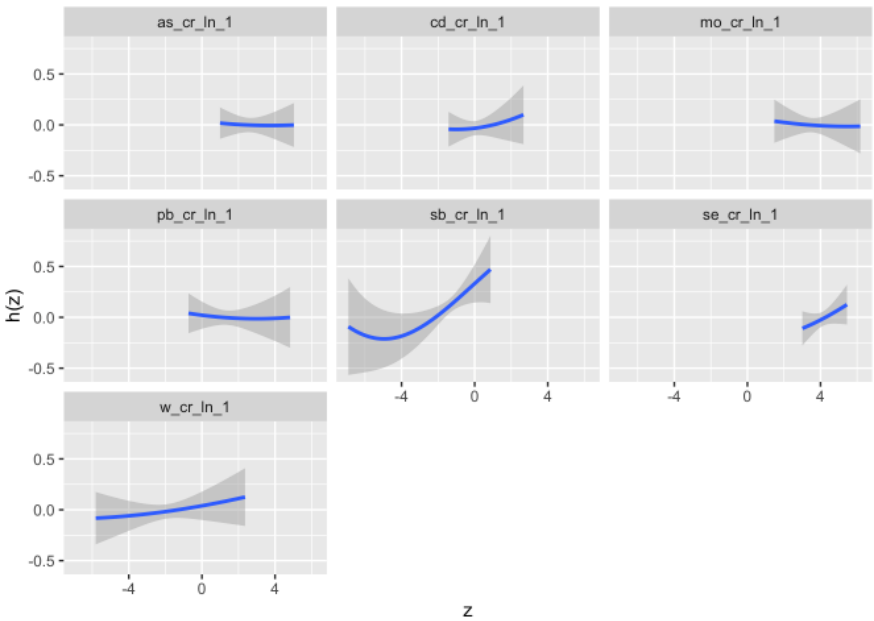

Figure 6 Predictor-response function for the first time-varying confounder, compares the increase in waist circumference at the second visit when a single pollutant is at the 75th versus 25th percentile, for male participants

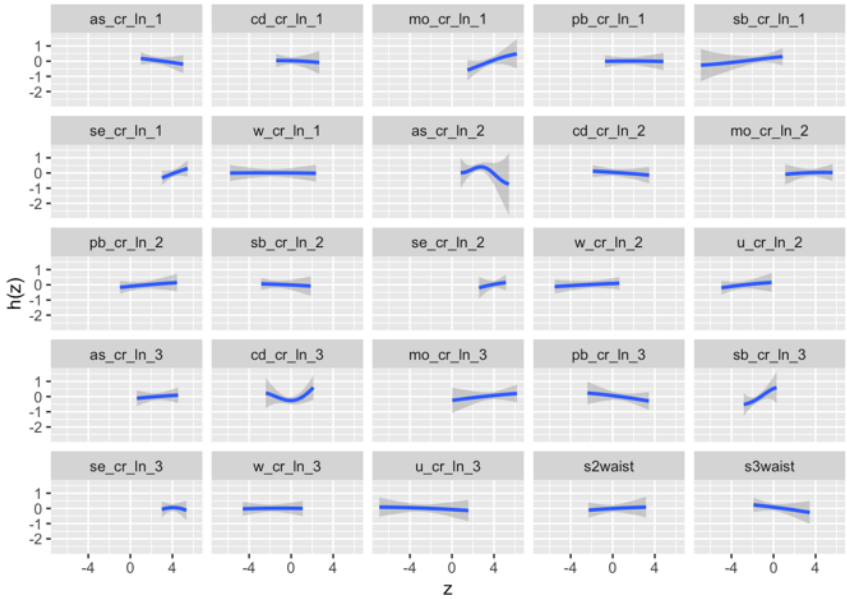

Figure 7 Predictor-response function for the outcome (blood glucose), compares the increase in glucose when a single pollutant is at the 75th versus 25th percentile, for male participants

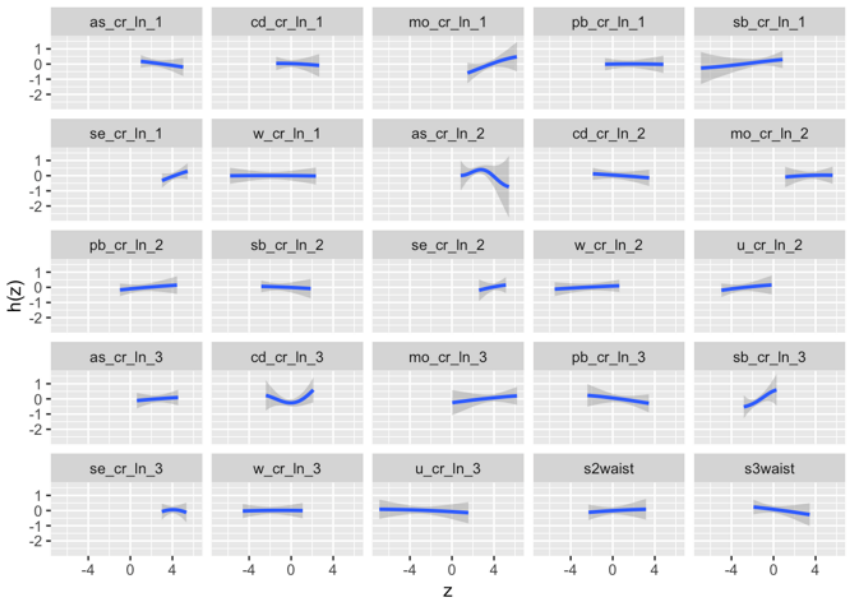

Figure 8 Predictor-response function for the outcome (blood glucose), compares the increase in glucose when a single pollutant is at the 75th versus 25th percentile, for male participants